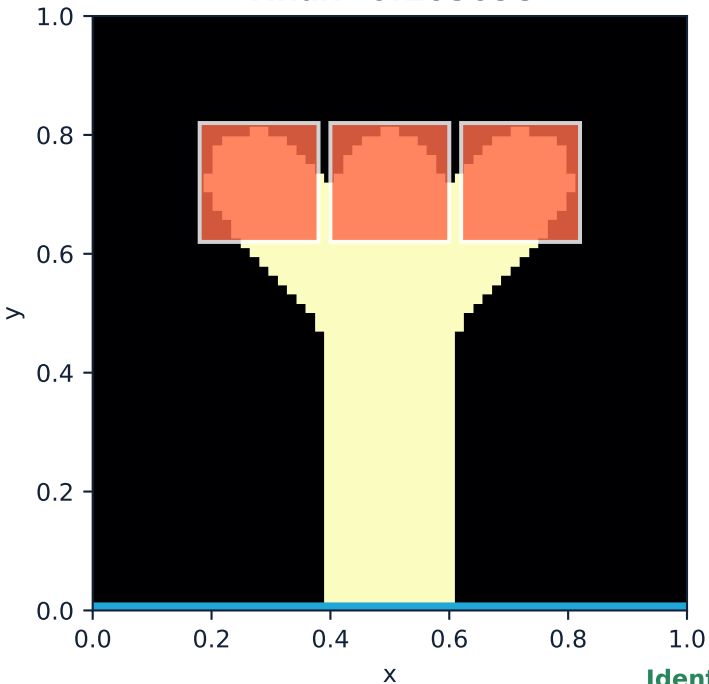


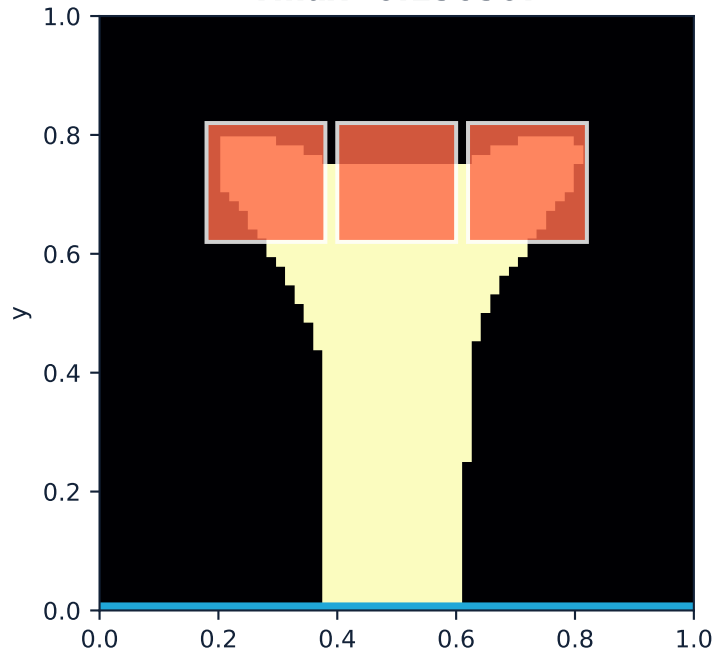
# Three strategies, one thermal objective

The neural rule can represent a topology competitive with direct pixel optimization

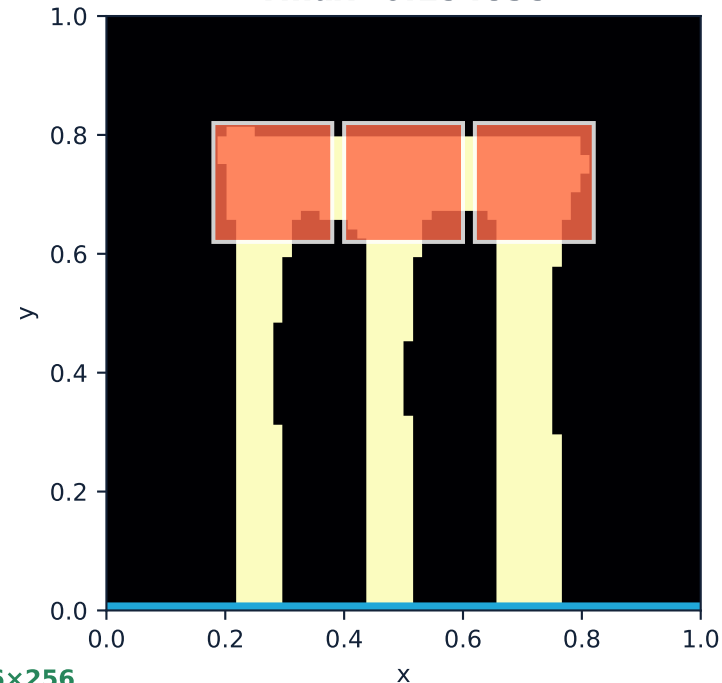
**Parametric tree**  
**Tmax=0.165098**



**Pixel WaveForge**  
**Tmax=0.156507**



**Pure NCA (best seed)**  
**Tmax=0.154835**



Identical strict-binary material budget  $\approx 25\%$  • independent SciPy 256x256